

Ryan Hoque

<https://ryanhoque.github.io> | SF Bay Area, CA

ACADEMIA

UC Berkeley — Ph.D., Computer Science

AUGUST 2020 - MAY 2024

- Advisor: Ken Goldberg
- Dissertation Committee: Ken Goldberg, Pieter Abbeel, Anca Dragan, Shuran Song
- Berkeley Artificial Intelligence Research (BAIR) Lab. Thesis: [Scalable Lifelong Imitation Learning for Robot Fleets](#)

UC Berkeley — M.S., Electrical Engineering and Computer Sciences

AUGUST 2019 - MAY 2020

- GPA: 4.00/4.00
- Advisor: Ken Goldberg
- Thesis: [Robotic Fabric Manipulation with Deep Imitation Learning and Reinforcement Learning in Simulation](#)

UC Berkeley — B.S., Electrical Engineering and Computer Sciences

AUGUST 2016 - MAY 2019

- GPA: 4.00/4.00

INDUSTRY

Meta, Burlingame, CA — Research Scientist

SEPTEMBER 2025 - PRESENT

- Robot learning research at Meta Robotics Studio (within Meta Reality Labs)

Apple, Cupertino, CA — Research Scientist

AUGUST 2024 - SEPTEMBER 2025

- Fundamental and applied robot learning research with Apple Special Projects

NVIDIA, Seattle, WA — Research Scientist Intern

MAY 2023 - SEPTEMBER 2023

- Imitation learning research with NVIDIA's Seattle Robotics Lab

PREPRINTS AND PUBLICATIONS (Reverse Chronological)

1. **R. Hoque***, P. Huang*, D. Yoon*, M. Sivapurapu, J. Zhang. EgoDex: Learning Dexterous Manipulation from Large-Scale Egocentric Video. 1st Workshop on Egocentric Perception and Action for Robot Learning at *Robotics: Science and Systems (RSS)*, 2025. **Best Paper Award**.
2. R. Qiu, S. Yang, X. Cheng, C. Chawla, J. Li, T. He, G. Yan, D. Yoon, **R. Hoque**, L. Paulsen, G. Yang, J. Zhang, S. Yi, G. Shi, X. Wang. Humanoid Policy ~ Human Policy. *Conference on Robot Learning (CoRL)*, 2025.
3. N. Nechyporenko*, **R. Hoque***, C. Webb, M. Sivapurapu, J. Zhang. ARMADA: Augmented Reality for Robot Manipulation and Robot-Free Data Acquisition. *Arxiv preprint 2412.10631*, 2024.
4. **R. Hoque**, A. Mandlekar, C. Garrett, K. Goldberg, D. Fox. IntervenGen: Interventional Data Generation for Robust and Data-Efficient Robot Imitation Learning. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.
5. Open X-Embodiment Collaboration, A. Padalkar, A. Pooley, A. Jain, A. Bewley, A. Herzog, A. Irpan, A. Khazatsky, A. Rai, A. Singh, A. Brohan, A. Raffin, A. Wahid, B. Burgess-Limerick, B. Kim, B. Schölkopf, B. Ichter, C. Lu, C. Xu, C. Finn, C. Xu, C. Chi, C. Huang, C. Chan, C. Pan, C. Fu, C. Devin, D. Driess, D. Pathak, D. Shah, D. Büchler, D. Kalashnikov, D. Sadigh, E. Johns, F. Ceola, F. Xia, F. Stulp, G. Zhou, G.S. Sukhatme, G. Salhotra, G. Yan, G. Schiavi, G. Kahn, H. Su, H. Fang, H. Shi, H. Amor, H. Christensen, H. Furuta, H. Walke, H. Fang, I. Mordatch, I. Radosavovic, I. Leal, J. Liang, J. Abou-Chakra, J. Kim, J. Peters, J. Schneider, J. Hsu, J. Bohg, J. Bingham, J. Wu, J. Wu, J. Luo, J. Gu, J. Tan, J. Oh, J. Malik, J. Tompson, J. Yang, J.J. Lim, J. Silvério, J. Han, K. Rao, K. Pertsch, K. Hausman, K. Go, K. Gopalakrishnan, K. Goldberg, K. Byrne, K. Oslund, K. Kawaharazuka, K. Zhang, K. Rana, K. Srinivasan, L.Y. Chen, L. Pinto, L. Tan, L. Ott, L. Lee, M. Tomizuka, M. Du, M. Ahn, M. Zhang, M. Ding, M.K. Srirama, M. Sharma, M. Kim, N. Kanazawa, N. Hansen, N. Heess, N.J. Joshi, N. Suenderhauf, N. Di Palo, N. Shafiullah, O. Mees, O. Kroemer, P.R. Sanketi, P. Wohlhart, P. Xu, P. Sermanet, P. Sundaresan, Q. Vuong, R. Rafailov, R. Tian, R. Doshi, R. Martín-Martín, R. Mendonca, R. Shah, **R. Hoque**, R. Julian, S. Bustamante, S. Kirmani, S. Levine, S. Moore, S. Bahl, S. Dass, S. Sonawani, S. Song, S. Xu, S. Haldar, S. Adebola, S. Guist, S. Nasiriany, S. Schaal, S. Welker, S. Tian, S. Dasari, S. Belkhale, T. Osa, T. Harada, T. Matsushima, T. Xiao, T. Yu,

- T. Ding, T. Davchev, T.Z. Zhao, T. Armstrong, T. Darrell, V. Jain, V. Vanhoucke, W. Zhan, W. Zhou, W. Burgard, X. Chen, X. Wang, X. Zhu, X. Li, Y. Lu, Y. Chebotar, Y. Zhou, Y. Zhu, Y. Xu, Y. Wang, Y. Bisk, Y. Cho, Y. Lee, Y. Cui, Y. Wu, Y. Tang, Y. Zhu, Y. Li, Y. Iwasawa, Y. Matsuo, Z. Xu, Z.J. Cui. Open X-Embodiment: Robotic Learning Datasets and RT-X Models. *IEEE International Conference on Robotics and Automation*, 2024. **Best Conference Paper Award**.
6. G. Datta*, **R. Hoque***, A. Gu, E. Solowjow, K. Goldberg. IIFL: Implicit Interactive Fleet Learning from Heterogeneous Human Supervisors. *Conference on Robot Learning (CoRL)*, 2023.
 7. S. Sharma, K. Shivakumar, H. Huang, L. Chen, **R. Hoque**, B. Ichter, K. Goldberg. Semantic Mechanical Search with Large Vision and Language Models. *Conference on Robot Learning (CoRL)*, 2023.
 8. K. Chen, **R. Hoque**, K. Dharmarajan, E. Lontop, S. Adebola, J. Ichnowski, J. Kubiawicz, K. Goldberg. FogROS2-SGC: A ROS2 Cloud Robotics Platform for Secure Global Connectivity. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
 9. J. Kerr, H. Huang, A. Wilcox, **R. Hoque**, J. Ichnowski, R. Calandra, K. Goldberg. Learning Self-Supervised Representations from Vision and Touch for Active Sliding Perception of Deformable Surfaces. *Robotics: Science and Systems (RSS)*, 2023.
 10. **R. Hoque**, L.Y. Chen, S. Sharma, K. Dharmarajan, B. Thananjeyan, P. Abbeel, K. Goldberg. Fleet-Dagger: Interactive Robot Fleet Learning with Scalable Human Supervision. *Conference on Robot Learning (CoRL)*, 2022. **Oral Presentation** (6.5% of papers).
 11. **R. Hoque***, K. Shivakumar*, S. Aeron, G. Deza, A. Ganapathi, A. Wong, J. Lee, A. Zeng, V. Vanhoucke, K. Goldberg. Learning to Fold Real Garments with One Arm: A Case Study in Cloud-Based Robotics Research. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.
 12. S. Sharma*, E. Novoseller*, V. Viswanath, Z. Javed, R. Parikh, **R. Hoque**, A. Balakrishna, D.S. Brown, K. Goldberg. Learning Switching Criteria for Sim2Real Transfer of Robotic Fabric Manipulation Policies. *IEEE Conference on Automation Science and Engineering (CASE)*, 2022.
 13. **R. Hoque***, D. Seita*, A. Balakrishna, A. Ganapathi, A. Tanwani, N. Jamali, K. Yamane, S. Iba, K. Goldberg. VisuoSpatial Foresight for Physical Sequential Fabric Manipulation. *Autonomous Robots* 46, 175–199 (2022).
 14. **R. Hoque**, A. Balakrishna, E. Novoseller, A. Wilcox, D. Brown, K. Goldberg. ThriftyDagger: Budget-Aware Novelty and Risk Gating for Interactive Imitation Learning. *Conference on Robot Learning (CoRL)*, 2021. **Oral Presentation** (6.5% of papers).
 15. **R. Hoque**, A. Balakrishna, C. Putterman, M. Luo, D. Brown, D. Seita, B. Thananjeyan, E. Novoseller, K. Goldberg. LazyDagger: Reducing Context Switching in Interactive Imitation Learning. *IEEE Conference on Automation Science and Engineering (CASE)*, 2021.
 16. A. Ganapathi, P. Sundaresan, B. Thananjeyan, A. Balakrishna, D. Seita, J. Grannen, M. Hwang, **R. Hoque**, J. Gonzalez, N. Jamali, K. Yamane, S. Iba, K. Goldberg. Learning dense visual correspondences in simulation to smooth and fold real fabrics. *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.
 17. **R. Hoque***, D. Seita*, A. Balakrishna, A. Ganapathi, A. Tanwani, N. Jamali, K. Yamane, S. Iba, K. Goldberg. VisuoSpatial Foresight for Multi-Step, Multi-Task Fabric Manipulation. *Robotics: Science and Systems (RSS)*, 2020.
 18. D. Seita, A. Ganapathi, **R. Hoque**, M. Hwang, E. Cen, A. Tanwani, A. Balakrishna, B. Thananjeyan, J. Ichnowski, N. Jamali, K. Yamane, S. Iba, J. Canny, K. Goldberg. Deep Imitation Learning of Sequential Fabric Smoothing from an Algorithmic Supervisor. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020.
 19. A. Ganapathi, P. Sundaresan, B. Thananjeyan, A. Balakrishna, D. Seita, **R. Hoque**, J. Gonzalez, K. Goldberg. MMGSD: Multi-Modal Gaussian Shape Descriptors for Correspondence Matching in 1D and 2D Deformable Objects. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) Workshop on Managing Deformation*, 2020.

HONORS AND AWARDS

Best Paper Award, EgoAct Workshop at Robotics: Science and Systems (RSS) 2025	JUNE 2025
Best Conference Paper Award, IEEE International Conference on Robotics and Automation (ICRA) 2024	MAY 2024
Oral Presentation, Conference on Robot Learning (CoRL) 2022 in Auckland, New Zealand	FALL 2022
Oral Presentation, Conference on Robot Learning (CoRL) 2021 in London, UK	FALL 2021
UC Berkeley Quantedge Award for Academic Excellence	FALL 2018
Best Web App at Cal Hacks, the World's Largest Collegiate Hackathon	NOV 2016 & 2017
UC Berkeley Edward Frank Kraft Award	FALL 2016
Outstanding Academic Achievement in Mathematics – top math student in graduating class of 400+	MAY 2016
National Merit Scholarship – Merit-based award for ~7,000 students out of 1.5 million	MAY 2016

PROFESSIONAL SERVICES

Organizing Committee for Conference on Robot Learning (CoRL) 2022

Program Committee for Robot Learning Workshop at NeurIPS 2023

UC Berkeley EECS Graduate Admissions Committee 2023

Teaching Assistant for EE 16B: Designing Information Devices and Systems II, Spring 2024

Head Teaching Assistant for IEOR 375: GSI Proseminar on Teaching Engineering, Fall 2023

Program Committee for Safe and Robust Control Workshop at NeurIPS 2021

Volunteer Manager of Berkeley AI Research Social Media (150K+ Followers)

Keynote Speaker for Berkeley HKN EECS Day 2022 (outreach event for 100+ high school students)

AI Student Lead for UC Berkeley EECS PhD Visit Days for prospective students in 2022 and 2023

Mentor for the BAIR Undergraduate Mentoring Program in Spring 2022

Reviewer for Robotics: Science and Systems (RSS), Conference on Robot Learning (CoRL), IEEE International Conference on Robotics and Automation (ICRA), IEEE International Conference on Intelligent Robots and Systems (IROS), Applied

Mathematical Modelling, Neural Information Processing Systems (NeurIPS), IEEE Transactions on Robotics (T-RO), IEEE Robotics and Automation Letters (RA-L), ACM Transactions on Human-Robot Interaction (THRI), International Journal of Robotics Research (IJRR)